

FINAL PERFORMANCE TEST REPORT

DemoBlaze Web Application

<https://www.demoblaze.com>

Baseline | Load | Peak | Stress | Endurance

Field	Details
Report Version	1.0 — Final
Test Execution Date	16 March 2026
Application Under Test	DemoBlaze — https://www.demoblaze.com
Performance Tool	Apache JMeter 5.6.3
Test Environment	Windows 11 (local) / Ubuntu 22.04 (GitHub Actions)
Prepared By	AmaliTech Training Academy — Team 1
Overall Result	PARTIAL PASS — Baseline, Load, Peak, Stress passed. Endurance FAILED (3.75% error rate due to CDN rate limiting)

1. Executive Summary

This report presents the results of performance testing conducted on the DemoBlaze web application (www.demoblaze.com) using Apache JMeter 5.6.3. Testing was executed across five phases: Baseline (50 users), Load (150 users), Peak (300 users), Stress (500 users), and Endurance (100 users / 30 minutes).

The DemoBlaze application demonstrated consistent performance under Baseline, Load, Peak, and Stress test phases, with response times remaining well below the 2,000ms SLA threshold. Notably, average response times improved as concurrent user counts increased, which is characteristic of a CDN-backed static application with strong caching.

However, the Endurance test FAILED the 1% error rate SLA. A 3.75% error rate was recorded, caused by GitHub Pages CDN rate limiting under sustained 100-user load over approximately 30 minutes. This is an environmental constraint rather than an application defect, but it is recorded as a SLA breach.

1.1 Key Findings

Baseline, Load, Peak, and Stress tests all completed with 0% error rate.

Average response times ranged from 368ms to 392ms across primary phases, well within the 2,000ms SLA.

The application demonstrated stable scalability from 50 to 500 concurrent users with no breaking point found.

Endurance test FAILED: 225 errors out of 6,000 samples (3.75%) due to GitHub Pages CDN rate limiting activating after approximately 3 minutes of sustained load.

Maximum response time of 324,302ms recorded in Endurance — caused by requests queued and then timed out by the rate limiter.

GitHub Actions CI/CD pipeline successfully executed Baseline, Load, and Stress tests with email and Slack notification delivery.

1.2 Overall Assessment

Test Phase	Users	Result	Notes
Baseline	50	PASS	0 errors, Avg 392ms, all SLAs met

Test Phase	Users	Result	Notes
Load	150	PASS	0 errors, Avg 382ms, throughput 8.7 req/s
Peak	300	PASS	0 errors, Avg 368ms, throughput 11.9 req/s
Stress	500	PASS	0 errors, Avg ~400ms — no breaking point found
Endurance	100	FAIL	3.75% error rate — exceeds 1% SLA. CDN rate limiting triggered after ~3 min

2. Test Environment

Component	Details
Application Under Test	DemoBlaze — https://www.demoblaze.com
Hosting Infrastructure	GitHub Pages (CDN-backed static hosting)
Protocol	HTTPS (TLS)
Performance Tool	Apache JMeter 5.6.3
JVM	OpenJDK 11
Load Generator — Local	Windows 11, Intel CPU, 16GB RAM
Load Generator — CI/CD	GitHub Actions ubuntu-latest runner
Test Execution Date	16 March 2026
Test Start Times	Baseline: 12:54 CAT Load: 13:14 CAT Peak: 13:22 CAT
Repository	github.com/giftburabyo-a11y/swaglab-performance-tests

3. Test Scenarios Executed

Each virtual user session executed the following transaction sequence. Gaussian Random Timers were applied between requests to simulate realistic user think times. HTTP 200 response code assertions were configured on every request.

Step	Transaction Name	Method	URL	Think Time
1	Load Home Page	GET	/index.html	2000ms ± 1000ms
2	View Product Detail (Samsung S6)	GET	/prod.html?idp_=1	2000ms ± 1000ms
3	Browse Another Product (Nokia)	GET	/prod.html?idp_=2	2000ms ± 1000ms
4	View Cart Page	GET	/cart.html	1500ms ± 500ms
5	Reload Home Page	GET	/index.html	1000ms ± 500ms
6	View Cart Again	GET	/cart.html	1000ms ± 500ms

4. Detailed Test Results

4.1 Baseline Test — 50 Concurrent Users

The baseline test was executed with 50 virtual users, a 60-second ramp-up period, and 3 loops per user. This test establishes the performance benchmark under light load conditions.

Metric	Value
Test Configuration	50 users Ramp-up: 60s Loops: 3
Total Samples	900
Test Duration	4 minutes 14 seconds
Average Response Time	392ms
Minimum Response Time	248ms
Maximum Response Time	1,493ms
Throughput	3.5 requests/second
Error Count	0
Error Rate	0.00%
SLA Status (< 2000ms avg)	PASS
SLA Status (< 1% error rate)	PASS

4.1.1 Observations

Response times were stable throughout the test with no upward trend over time. The maximum response time of 1,493ms was observed on the initial home page load, likely due to DNS resolution and TCP connection setup for the first request per user. Throughput of 3.5 req/s is appropriate for 50 users with think times applied.

4.2 Load Test — 150 Concurrent Users

The load test was executed with 150 virtual users, a 120-second ramp-up period, and 3 loops per user. This test validates application performance under normal expected production traffic.

Metric	Value
Test Configuration	150 users Ramp-up: 120s Loops: 3
Total Samples	2,700
Test Duration	5 minutes 12 seconds
Average Response Time	382ms
Minimum Response Time	245ms
Maximum Response Time	4,493ms
Throughput	8.7 requests/second
Error Count	0
Error Rate	0.00%
SLA Status (< 2000ms avg)	PASS
SLA Status (< 1% error rate)	PASS

4.2.1 Observations

Average response time decreased by 10ms compared to baseline (382ms vs 392ms), demonstrating that the application benefits from connection reuse and caching at higher concurrency. A maximum of 4,493ms was observed on a single request — an outlier likely caused by a network spike, as P95 remained within SLA bounds. Throughput scaled from 3.5 to 8.7 req/s with no degradation at 3x the baseline user load.

4.3 Peak Load Test — 300 Concurrent Users

The peak load test was executed with 300 virtual users, a 180-second ramp-up period, and 2 loops per user. This test evaluates application behavior under maximum expected traffic.

Metric	Value
Test Configuration	300 users Ramp-up: 180s Loops: 2
Total Samples	3,600
Test Duration	5 minutes 4 seconds

Metric	Value
Average Response Time	368ms
Minimum Response Time	242ms
Maximum Response Time	1,472ms
Throughput	11.9 requests/second
Error Count	0
Error Rate	0.00%
SLA Status (< 2000ms avg)	PASS
SLA Status (< 1% error rate)	PASS

4.3.1 Observations

Average response time continued to decrease (368ms) compared to baseline and load tests, confirming CDN caching effects at high concurrency. Maximum response time of 1,472ms was lower than the load test maximum, indicating stability under peak load. The application handled 6x the baseline user load with zero errors and improved response times.

4.4 Stress Test — 500 Concurrent Users

The stress test was executed with 500 virtual users, a 120-second ramp-up period, and 1 loop per user. This test is designed to identify the application's breaking point.

Metric	Value
Test Configuration	500 users Ramp-up: 120s Loops: 1
Total Samples	3,000
Test Duration	~5 minutes
Average Response Time	366ms
Minimum Response Time	246ms
Maximum Response Time	1,998ms
Throughput	16.1 requests/second

Metric	Value
Error Count	0
Error Rate	0.00%
SLA Status	Informational — stress test is not a pass/fail gate

4.4.1 Observations

No breaking point was found at 500 concurrent users. The application maintained sub-2000ms response times and 0% error rate, consistent with all prior phases. The CDN-backed static architecture absorbs concurrency well within a single test run duration.

4.5 Endurance Test — 100 Concurrent Users / 30 Minutes — FAILED

The endurance test was executed with 100 virtual users, a 120-second ramp-up period, and 20 loops per user, targeting approximately 30 minutes of sustained load. This test evaluates sustained performance and detects memory leaks or progressive degradation.

Metric	Value
Test Configuration	100 users Ramp-up: 120s Loops: 20 Target: ~30 min
Total Samples	6,000
Test Duration	12 minutes 32 seconds (rate-limited before 30 min target)
Average Response Time	199ms
Minimum Response Time	0ms (timeout/connection refusal errors)
Maximum Response Time	324,302ms (rate-limit queue timeout)
Throughput	8.0 requests/second
Error Count	225
Error Rate	3.75%
SLA Status (< 1% error rate)	FAIL — 3.75% exceeds 1% threshold

4.5.1 Root Cause Analysis

The endurance test error rate of 3.75% exceeded the 1% SLA threshold. Root cause analysis identified GitHub Pages CDN rate limiting as the cause. DemoBlaze is hosted on GitHub Pages, which enforces rate limits on excessive automated requests. Under sustained 100-user concurrent load with 20 loops, the infrastructure triggered rate limiting after approximately 3 minutes of continuous traffic.

The maximum response time of 324,302ms (over 5 minutes) confirms that some requests were held in a queue by the rate limiter before timing out rather than receiving an immediate error. The 0ms minimum response times correspond to requests that received immediate connection refusal once rate limiting fully activated.

4.5.2 Timeline Analysis

Time Interval	Avg Response (ms)	Error Rate	Throughput (req/s)	Status
0 – 2 min	~140ms	0.00%	~13	Normal — all requests passing
2 – 3 min	~125ms	0.00%	~21	Peak throughput — CDN cache fully warm
3 – 4 min	~277ms	32.75%	~11	Rate limiting activated — SLA BREACHED
4 – 8 min	~120ms	0.00%	~21	Rate limiting relaxed — recovery observed
8 – 12 min	~7,434ms	2.22%	~0.1	Final loops timing out — connection pool exhaustion

5. Comparative Analysis

5.1 Response Time Comparison Across All Test Phases

Test Phase	Users	Samples	Avg (ms)	Min (ms)	Max (ms)	Throughput	Errors	Error Rate
Baseline	50	900	392	248	1,493	3.5 req/s	0	0.00%
Load	150	2,700	382	245	4,493	8.7 req/s	0	0.00%
Peak	300	3,600	368	242	1,472	11.9 req/s	0	0.00%
Stress	500	3,000	366	246	1,998	16.1 req/s	0	0.00%
Endurance	100	6,000	199	0	324,302	8.0 req/s	225	3.75% FAIL

5.2 Scalability Analysis

Response times improved as user count increased from 50 to 300 users (392ms to 368ms), demonstrating positive scalability likely due to CDN edge caching becoming more efficient as more requests hit cached content.

Error rate remained at 0.00% across all four primary test phases (Baseline through Stress), indicating no functional degradation under load.

Throughput scaled proportionally with user count: 3.5 req/s at 50 users, 8.7 at 150, 11.9 at 300, and 16.1 at 500 — a near-linear scaling curve.

No breaking point was found within the tested load levels. The endurance failure was infrastructure-level rate limiting, not an application failure.

5.3 SLA Compliance Summary

SLA Requirement	Threshold	Baseline	Load	Peak	Endurance	Met?
Average Response Time	< 2,000ms	392ms	382ms	368ms	199ms (excluding timeouts)	YES — primary phases

SLA Requirement	Threshold	Baseline	Load	Peak	Endurance	Met?
Error Rate	< 1%	0.00%	0.00%	0.00%	3.75%	NO — Endur ance FAILE D
Min Response Time	< 500ms	248ms	245ms	242ms	0ms (errors)	YES — primar y phase s
Throughput	>= 5 req/s	3.5 req/s	8.7 req/s	11.9 req/s	8.0 req/s	YES — Load+
Endurance Stability	< 1% errors over 30 min	N/A	N/A	N/A	3.75%	NO — FAILE D

6. Bottleneck Analysis

6.1 Identified Bottleneck: GitHub Pages Rate Limiting

Attribute	Details
Bottleneck Type	Infrastructure-level rate limiting by GitHub Pages CDN
Affected Test Phase	Endurance (100 users / 30 minutes)
Symptom	Error rate spike to 32.75% at ~3 minutes; partial recovery at ~4 minutes; final connection pool exhaustion at 8–12 minutes
Root Cause	GitHub Pages enforces automated request rate limits to protect shared hosting infrastructure. Sustained automated load from 100 users over 20 loops triggered this threshold.
Impact	3.75% overall error rate in endurance test — above the 1% SLA threshold. Test terminated at 12 min 32 sec instead of 30 min target.
Severity	Medium — only affects extended automated load testing, not real user traffic
Recommendation	This is an environmental constraint, not a DemoBlaze application defect. For production systems on dedicated infrastructure, endurance testing should be re-evaluated against a locally hosted or dedicated instance.

6.2 Observation: Max Response Time Spikes

Attribute	Details
Observation	Single request max response times exceed average by 10–12x in Load; by orders of magnitude in Endurance
Affected Phases	Load (max 4,493ms), Endurance (max 324,302ms)
Root Cause	CDN cache misses on first request per user per session; Endurance spikes caused by rate-limit queue timeouts
Impact	Low in primary phases — isolated outliers. Average and P90 metrics remain within SLA. High in Endurance due to rate limiting.
Recommendation	Monitor P90 and P95 metrics rather than maximum values for SLA reporting. Add P90 threshold checks to CI pipeline.

7. CI/CD Pipeline Results

The GitHub Actions performance pipeline was configured and executed successfully. The pipeline is located at `.github/workflows/jmeter-performance-pipeline.yml` in the repository `github.com/giftburabyo-a11y/swaglab-performance-tests`.

7.1 Pipeline Execution Summary

Job	Users	Status	Artifacts Generated
Baseline Test (50 Users)	50	PASSED	baseline-html-report, baseline-jtl
Load Test (150 Users)	150	PASSED	load-html-report
Stress Test (500 Users)	500	PASSED	stress-html-report
Test Summary	N/A	PASSED	GitHub Step Summary with results table

7.2 Notification Delivery

Email notifications are configured via `GMAIL_ADDRESS`, `GMAIL_APP_PASS`, and `REPORT_EMAIL` secrets in GitHub repository settings. When configured, a full HTML report is delivered to the specified recipient after each pipeline run.

Slack notifications are configured via `SLACK_WEBHOOK_URL` secret. When configured, a structured Slack message block is sent to the configured channel with test metrics, run details, and artifact links.

8. Recommendations

8.1 Immediate Actions

Accept current performance results for primary phases. The application meets all Baseline, Load, Peak, and Stress SLAs. No immediate action is required for these phases.

Document the endurance rate limiting as an environmental constraint rather than an application defect. Update the test plan to note this limitation for future reference and re-classify the endurance SLA as non-applicable for shared CDN hosting.

Add P90/P95 threshold checks to the CI pipeline. The current pipeline checks error rate only. Adding P90 response time checks would provide earlier warning of response time regressions.

8.2 Medium-Term Improvements

Test against a dedicated server for more realistic endurance testing. A locally hosted or dedicated instance of the application would remove CDN rate limiting as a variable and produce more meaningful endurance results.

Expand CI/CD pipeline to include the endurance test. Currently the pipeline runs Baseline, Load, and Stress. Adding a scheduled nightly endurance test against a dedicated server would provide continuous visibility into sustained performance.

Parameterize SLA thresholds. Move the 1% error rate and 2000ms response time thresholds from hardcoded values to pipeline environment variables to allow easy adjustment per release.

8.3 Long-Term Monitoring

Implement continuous performance monitoring using tools such as Grafana and Prometheus if the application is migrated to a dedicated server environment.

Track performance trends across multiple test runs using the JTL result files. Comparing P90 response times across weekly runs would detect gradual performance degradation before it becomes visible to users.

9. Conclusion

The DemoBlaze web application was evaluated across five performance test phases. It demonstrated excellent and consistent performance under Baseline, Load, Peak, and Stress conditions, meeting and exceeding all primary SLAs with 0% error rate across 9,600 total requests.

The most significant positive finding is that average response time improved as concurrent user load increased from 50 to 300 users (392ms to 368ms), confirming an efficiently cached CDN-optimized architecture. The application processed 3,600 peak-load requests and 3,000 stress-level requests with zero errors, confirming its ability to handle real-world traffic levels.

The Endurance test FAILED the 1% error SLA, recording 225 errors out of 6,000 samples (3.75%). Root cause was confirmed as GitHub Pages CDN rate limiting under sustained automated load — not an application defect. The test also terminated at 12 minutes 32 seconds rather than the 30-minute target due to connection pool exhaustion caused by the rate limiter. This is recorded as a conditional finding rather than an application risk for real user traffic.

The automated CI/CD pipeline has been successfully implemented using GitHub Actions, providing continuous performance monitoring on every code commit with email and Slack notification capabilities.

9.1 Final Verdict by Test Phase

Test Phase	Users	Verdict	Justification
Baseline	50	PASS	0 errors, Avg 392ms — all SLAs met
Load	150	PASS	0 errors, Avg 382ms — scales well from baseline
Peak	300	PASS	0 errors, Avg 368ms — best avg response across all phases
Stress	500	PASS	0 errors, Avg 366ms, 16.1 req/s — no breaking point found
Endurance	100 / 30 min	FAIL	3.75% error rate — exceeds 1% SLA. CDN rate limiting caused 225 errors and test terminated at 12 min 32 sec instead of 30 min target
CI/CD Pipeline	N/A	PASS	All jobs succeeded — Baseline, Load, Stress. Notifications configured.